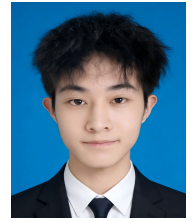




中山大學
SUN YAT-SEN UNIVERSITY

Yongxue Xu (Jerry)

✉ xuyx85@mail2.sysu.edu.cn
🌐 jerrysnow.me 📺 IamJerryXu
🎓 Google Scholar 🔗 LinkedIn



👤 Summary

Undergraduate researcher at Sun Yat-sen University working on video understanding and generation, multimodal foundation models, world models, and 4D scene understanding. I am particularly interested in models that connect perception, generation, and interaction, and I am currently seeking industry research internships in these areas.

🎓 Education

Sun Yat-sen University (SYSU) *Project 985 / 211 / Double First-Class* Sep. 2023 – Jun. 2027

B.Eng. in Intelligent Science and Technology, School of Intelligent Systems Engineering

GPA: 3.91 / 4.0 | **Major GPA:** 3.94 / 4.0 (*Major Required + Major Elective*)

🔧 Skills

Research	Video understanding and generation; multimodal foundation models; world models; 4D scene understanding; world-action models.
Programming	Python, C/C++, MATLAB; experienced with PyTorch.
Languages	Chinese (native); English (CET-6, academic reading and writing).

🔬 Research Experience

LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue

- **Overview:** Built a multi-view temporal benchmark spanning three domains and proposed **LMM-Track4D** for object-level 4D reasoning. Trajectory-grounded dialogue, a streaming [TRK] state token, and trajectory decoding support both textual QA and 3D trajectory prediction.
- **Responsibilities:** Led the technical roadmap, model design, and main experiments; contributed to benchmark construction.
- **Outcome:** Submitted to **NeurIPS 2026** as **co-first author**.

Learning to Reason with Persistent Object States for Video Instance Segmentation

- **Overview:** Proposed **POSReasoner**, a trainable plug-and-play framework that reasons over persistent object states. A sparse state-observation graph and Propose-Verify procedure resolve competing associations and selectively retain, update, or reactivate states while the host model remains frozen.
- **Responsibilities:** Led the technical roadmap, architecture, implementation, experiments, and manuscript development.
- **Outcome:** Preprint; **co-first author**.

Graph Domain Adaptation Does Not End with Representation Learning

- **Overview:** Proposed **EviGDA**, pairing an entropy-aligned graph-aware expert with an independent graph-free expert. Post-training fusion exploits complementary evidence without learned routing or target pseudo-labels.
- **Outcome:** State-of-the-art results on ten datasets and 16 transfer tasks; preprint; **co-first author**.

Robust Competitive Influence Maximization on Multilayer Networks

- **Overview:** Proposed a diffusion-aware, role-guided evolutionary framework combining cross-layer alignment, role constraints, dynamic robustness evaluation, and multi-task evolutionary search.
- **Responsibilities:** Defined the roadmap; designed core modules and experiments; led patent filing and manuscript development.
- **Outcome:** **1 patent pending**; manuscript under review at **IEEE TNSE** as **first author**.

Critical Node Identification in Urban Road Networks via Deep Reinforcement Learning

- **Overview:** Evaluated deep and reinforcement learning methods on real urban road networks, using GNNs for topology representation and DQN-based agents for adaptive critical-node decisions. Contributed dataset construction, baseline reproduction, and visualization.
- **Outcome:** **Third author**; paper received the **Best Paper Award** at **GBCESC 2025**.

Selected Publications

MANUSCRIPTS AND PREPRINTS

LMM-Track4D: Eliciting 4D Dynamic Reasoning in LMMs via Trajectory-Grounded Dialogue

Chaoyue Li*, **Yongxue Xu***, Jie Feng*, and Jiayu Ding[†]

Submitted to NeurIPS 2026

2026

Learning to Reason with Persistent Object States for Video Instance Segmentation

Yongxue Xu*, Boxue Yang*, Ziqian Liu, Shaoqiu Zhang, and Linfeng Zhang[†]

Preprint

2026

Graph Domain Adaptation Does Not End with Representation Learning

Yongxue Xu*, Ziqian Liu*, Enze Zhang, and Maolin Wang[†]

Preprint

2026

Solving the Robust Influence Maximization Problem in Competitive Multilayer Networks via a Diffusion-Aware Role-Guided Evolutionary Approach

Yongxue Xu, Ziqian Liu, Yongqing Huang, and Shuai Wang[†]

Under review at IEEE Transactions on Network Science and Engineering

2026

CONFERENCE PAPER

Identifying Critical Nodes with Deep Learning and Reinforcement Learning: A Case Study on Urban Road Networks

Zhonghan Liu, Wenshuo Liu, **Yongxue Xu**, Ziqian Liu, Yichen Liu, and Shuai Wang[†]

GBCESC 2025, Best Paper Award

2025

* Equal contribution. [†] Corresponding author.

Research Initiatives & Collaborations

- Collaborate with researchers at **HKUST**, the **EPIC Lab at SJTU SAI**, and **Westlake University** on video generation, world-action modeling, and multimodal spatiotemporal understanding.
- Co-organize **InkMind.AI**, a small research group exploring multimodal understanding and generation. We also study vibe-coding workflows for recursive self-improvement and automated research, with the goal of making AI-assisted research more precise, efficient, and insight-driven.

Awards & Competitions

Chinese Mathematics Competition (CMC), Guangdong Region — Second Prize

Dec. 2024

Mathematical Contest in Modeling (MCM), USA — Honorable Mention

Feb. 2025

National Undergraduate Smart Car Competition, South China Region — Second Prize

Aug. 2025

Interests

Table tennis, music, and nature travel.